Software Engineer

Data & Cloud (AWS) | Backend & Data Systems |

Christian J. Medina Díaz

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Professional Summary

Software Engineer with experience building data-driven systems and cloud-based data pipelines. Skilled in backend development, relational databases, and data engineering concepts, with hands-on experience working with large-scale datasets (1M+ records). Proficient in Python, SQL, and PostgreSQL, with a strong foundation in system design, data modeling, and query optimization. Currently working with cloud technologies to design scalable data solutions, from ingestion to visualization. Passionate about building efficient, scalable systems and continuously improving through hands-on problem solving in real-world environments.

Technical Skills

Languages: Python, Java, SQL, JavaScript, C/C++
Backend Development: Flask, RESTful API Design, Authentication & Session Management
Cloud & Data: AWS (S3, Athena, QuickSight), Data Pipelines, Data Processing
Database & Data Engineering: PostgreSQL, Relational Modeling, Schema Design, Data Normalization (3NF), Query Optimization, Indexing
Systems & Architecture: Clean Architecture, SOLID Principles, SDLC, Agile/Scrum, Role-Based Access Control (RBAC), Requirements Analysis
Analytics & Reporting: Data Validation, Performance Metrics Tracking, Matplotlib, Streamlit Dashboards
Tools & Platforms: Git, GitHub, Bitbucket, Heroku, Supabase
Operating Systems: Linux, Windows, macOS

Professional Experience

Software Engineer (Independent Contractor) - Innovatio Software Solutions
Puerto Rico | 2026 – Present
-Designing and implementing end-to-end data pipelines using cloud-based technologies.
-Working with large-scale datasets (1M+ records), focusing on performance, scalability, and cost optimization.
-Performing data ingestion, transformation, and querying within cloud data environments.
-Writing optimized SQL queries for analytics and reporting use cases.
-Building data visualizations and dashboards to support data-driven decision making.
-Applying data cleaning and validation techniques to ensure data quality and integrity.
-Collaborating with cross-functional teams in an Agile environment.

Professional Projects

Sports & Exercise Management System – Backend & Data Engineering Lead
-Designed and implemented 15+ RESTful API endpoints to manage users, workouts, and performance metrics.
-Modeled and normalized PostgreSQL schema (3NF), reducing data redundancy and improving data integrity.
-Implemented indexing and optimized SQL queries, improving reporting performance by ~30%.
-Translated functional requirements into backend validation logic, access control policies, and analytics workflows.
-Built reporting dashboards to visualize performance trends and support data-driven decision-making.
-Integrated LLM-powered chatbot enabling natural language querying of structured relational data.
-Deployed full-stack system to Heroku with environment configuration and database migrations.
Technologies: Python, Flask, PostgreSQL, SQL, Streamlit, Git

EconQuest – Role-Based Enterprise Learning Platform (Capstone Project)
-Architected modular backend components supporting multi-role access control (Admin, Student, Teacher).
-Led development of the Student module, implementing secure authentication, transactional database operations, and protected routing.
-Converted stakeholder requirements into structured relational database schemas and scalable backend workflows.
-Developed analytics dashboards to track engagement metrics across user segments.
-Collaborated within 25+ member Agile team delivering milestone-driven features under Scrum methodology.
Technologies: Python, Flask, PostgreSQL, SQL, Heroku, Git

RegiUPR – Agile University Enrollment System
-Contributed to development of a desktop-based course enrollment system within a 22-member Agile team.
-Implemented user registration, input validation, and account recovery workflows aligned with functional requirements.
-Designed backend validation logic to enforce business rules and maintain data consistency.
-Participated in sprint planning, backlog refinement, and version-controlled collaborative development.
Technologies: Python, PyQt5, Git

Gamified Gym – Scalable Application Architecture Project
-Collaborated in 27-member cross-functional team to design modular system architecture following Clean Architecture and SOLID principles.
-Implemented backend service integrations for authentication and role-based navigation control.
-Modeled domain entities to ensure separation of concerns between presentation, business logic, and data layers.
-Contributed to sprint deliverables, code reviews, and technical documentation.
Technologies: React Native, Supabase, JavaScript, Git

NBA Relational Database Analysis – Data Modeling Project
-Designed multi-season PostgreSQL relational database supporting complex statistical reporting.
-Applied normalization, foreign key constraints, and schema optimization to maintain referential integrity.
-Developed advanced SQL queries using joins, aggregations, subqueries, and performance-based metrics.
-Conducted structured data validation to ensure analytical accuracy.
Technologies: PostgreSQL, SQL, Python

Education

University of Puerto Rico – Mayagüez, PR
Bachelor of Science in Computer Science and Engineering
Graduated: *December 2025* | GPA: 3.2 / 4.0
Relevant Coursework: Databases, Software Engineering, Operating Systems, Algorithms, Networking

Additional Experience

Delivery Driver – UberEATS & DoorDash
Puerto Rico | 2020 – Present
-Independently managed high-volume delivery operations, optimizing routes and time-sensitive scheduling.
-Maintained performance metrics aligned with customer satisfaction and service reliability standards.
-Demonstrated accountability and problem-solving under operational constraints.